

Open Newton-Cotes Formulas

1. Midpoint Rule ($n = 0$)

Nodes: $x_{-1} = a, x_0 = a + h, x_1 = b$ with $h = \frac{b-a}{2}$.

Derivation: We fit a constant Lagrange polynomial $L_0(x)$ at x_0 :

$$f(x) = L_0(x)f(x_0) + \frac{f''(\xi(x))}{2!}(x - x_0)^2$$

Note: The f' term vanishes over symmetric $[x_{-1}, x_1]$.

$$\begin{aligned} \int_{x_{-1}}^{x_1} f(x)dx &= f(x_0) \int_{x_{-1}}^{x_1} 1 dx + \int_{x_{-1}}^{x_1} \frac{f''(\xi(x))}{2}(x - x_0)^2 dx \\ a_0 &= \int_{x_{-1}}^{x_1} 1 dx = x_1 - x_{-1} = 2h \end{aligned}$$

Error:

$$E = \frac{f''(\eta)}{2} \int_{x_{-1}}^{x_1} (x - x_0)^2 dx = \frac{f''(\eta)}{2} \int_{-h}^h t^2 dt = \frac{f''(\eta)}{2} \cdot \frac{2h^3}{3} = \frac{h^3}{3} f''(\eta)$$

Formula:

$$\int_a^b f(x)dx \approx 2hf(x_0), \quad E = \frac{h^3}{3} f''(\xi), \quad |E| \leq \frac{h^3}{3} \max |f''(x)|$$

Explanation: Uses the function value at the midpoint to create a rectangular approximation; exact for linear polynomials.

2. Two-Point Open Rule ($n = 1$)

Nodes: $x_{-1} = a, x_0, x_1, x_2 = b, h = \frac{b-a}{3}$.

Derivation: Integrate linear Lagrange polynomials $L_0(x), L_1(x)$:

$$a_0 = \int_{x_{-1}}^{x_2} \frac{x - x_1}{x_0 - x_1} dx = -\frac{1}{h} \int_{x_{-1}}^{x_2} (x - x_1) dx = -\frac{1}{2h} [(x_2 - x_1)^2 - (x_{-1} - x_1)^2] = \frac{3h}{2}$$

By symmetry, $a_1 = \frac{3h}{2}$.

Error:

$$E = \int_{x_{-1}}^{x_2} \frac{f''(\xi)}{2}(x - x_0)(x - x_1) dx = \frac{3h^3}{4} f''(\xi)$$

Formula:

$$\int_a^b f(x)dx \approx \frac{3h}{2} [f(x_0) + f(x_1)], \quad |E| \leq \frac{3h^3}{4} \max |f''(x)|$$

Explanation: Uses two internal points; linear fit; avoids problematic endpoints.

3. Three-Point Open Rule ($n = 2$)

Nodes: $x_{-1} = a, x_0, x_1, x_2, x_3 = b, h = \frac{b-a}{4}$.

Derivation: Integrate quadratic Lagrange basis $L_i(x)$:

$$a_0 = \int_{x_{-1}}^{x_3} \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} dx = \frac{8h}{3}, \quad a_1 = \int_{x_{-1}}^{x_3} \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} dx = -\frac{4h}{3}, \quad a_2 = a_0 = \frac{8h}{3}$$

Formula:

$$\int_a^b f(x)dx \approx \frac{4h}{3} [2f(x_0) - f(x_1) + 2f(x_2)]$$

Error:

$$E = \frac{14h^5}{45} f^{(4)}(\xi), \quad |E| \leq \frac{14h^5}{45} \max |f^{(4)}(x)|$$

Explanation: Open counterpart of Simpson's $1/3$; degree 3 exactness; negative central weight compensates curvature.

4. Four-Point Open Rule ($n = 3$)

Nodes: $x_{-1} = a, x_0, x_1, x_2, x_3, x_4 = b, h = \frac{b-a}{5}$.

Derivation: Integrate cubic Lagrange basis $L_i(x)$:

$$a_0 = a_3 = \frac{55h}{24}, \quad a_1 = a_2 = \frac{5h}{24}$$

Formula:

$$\int_a^b f(x)dx \approx \frac{5h}{24}[11f(x_0) + f(x_1) + f(x_2) + 11f(x_3)]$$

Error:

$$E = \frac{95h^5}{144}f^{(4)}(\xi), \quad |E| \leq \frac{95h^5}{144} \max |f^{(4)}(x)|$$

Explanation: Four internal nodes; cubic-order precision; effective for smooth functions; shares same derivative order as three-point rule.